

Study Guide, Exercises, and Sample Questions

CSE 4615: Wireless Networks

Islamic University of Technology (IUT)

Lecture 2: Overview of Wireless Access Networks

1. Topics Covered

- Internet overview: networks, internetworks, communication links, end systems, and routers.
- Protocol stack overview: Transport Layer (multiplexing, reliability, congestion control) and Network Layer (routing, forwarding, QoS).
- Network structure: network edge, network core, physical media (wired and wireless).
- Access networks: definition, role of the edge router, types (residential, institutional, mobile).
- Taxonomy of access networks by technology: Ethernet, WiFi-based, WiMAX-based, Cellular, Low PAN.
- Wired access networks: DSL (Digital Subscriber Line) and Cable Networks.
- Wireless access networks: classification, range comparison, and milestones in wireless communications.
- Wireless network components: wireless link, wireless host, base station, and network infrastructure.
- Infrastructure modes: infrastructure mode (single-hop) vs. ad hoc mode (infrastructure-less).
- Classification matrix: single-hop vs. multi-hop $\times$ infrastructure-based vs. infrastructure-less.
- Single-hop Infrastructure-based:
 - Cellular Networks (3G architecture, cell reuse, handoff, MTSO).
 - Wireless LANs (802.11b/a/g/n/ac/ax/be standards and data rates).
 - WiMAX (IEEE 802.16, last-mile broadband, triple-play services).
- Single-hop Infrastructure-less:
 - Bluetooth Networks (piconet, master-slave, FHSS, device states).
- Multi-hop Infrastructure-based:
 - Wireless Mesh Networks (structure, routing, self-healing, self-configuration, applications, challenges).
- Multi-hop Infrastructure-less:

- Wireless Ad-Hoc Networks (peer-to-peer, dynamic topology).
- Wireless Sensor Networks (energy constraints, Low PAN / IEEE 802.15, applications).
- Vehicular Ad-Hoc Networks (VANET): V2V, V2I, C2X communication, ITS applications.
- Delay Tolerant Networks (DTN): store-and-forward, challenges, applications.
- Comparison: Wireless Mesh Networks vs. Wireless Ad-Hoc Networks.
- Comparison: WSN, VANET, and DTN common characteristics.
- Challenges of C2X communication: channel conditions, networking, mobility, security.

2. Focus Areas

- Definition and role of access networks; how they connect end systems to the first router.
- Taxonomy of access networks – understanding the full classification tree by technology.
- Difference between wired (DSL, Cable, Ethernet) and wireless access networks; typical data rates.
- DSL: how voice and data share the same telephone line; role of DSLAM and splitter; upstream vs. downstream rates.
- Key components of a wireless network: wireless host, wireless link, base station, network infrastructure.
- Infrastructure mode vs. ad hoc mode: definition, examples, and when each is used.
- The four-quadrant classification: single-hop/multi-hop $\times$ infrastructure-based/infrastructure-less, with real examples in each quadrant.
- Cellular Networks: cell-based geographic reuse, role of MTSO, handoff, co-channel interference.
- Wireless LAN standards: frequency bands, data rates, and modulation techniques for 802.11b/a/g/n/ac/ax/be.
- WiMAX: IEEE 802.16, data rates, use as “last mile” alternative to DSL/Cable.
- Bluetooth: piconet topology, master-slave roles, FHSS at 1600 hops/sec, device states (Active, Parked, Standby).
- Wireless Mesh Networks (WMN): mesh router vs. mesh client roles, multi-hop routing, self-healing, self-configuration, key challenges (interference, scalability, QoS, security).
- Ad-Hoc Networks: peer-to-peer, no backbone, dynamic topology, energy and capacity constraints.

- Wireless Sensor Networks: energy as primary constraint, low PAN, data aggregation toward a centralized sink, application domains.
- Vehicular Ad-Hoc Networks (VANET): V2V and V2I communication, RSU role, ITS applications (AIM, platooning, eco-driving, cooperative perception).
- Delay Tolerant Networks (DTN): store-and-forward paradigm, bundle protocol, challenges (large delay, intermittent links, no ARQ).
- Comparison of WSN, VANET, and DTN in terms of infrastructure, topology dynamics, and self-organization.

3. How to Study These Topics

1. Re-draw the taxonomy tree of access networks from memory and place real-world examples in each branch.
2. Trace the path of a data packet from a home DSL modem to a server on the Internet, identifying each network element.
3. Study the 3G cellular architecture diagram; be able to label all components and explain handoff.
4. Review the Bluetooth piconet: draw master-slave connections, label parked and standby nodes, and state the FHSS parameters.
5. For WMNs: summarize each challenge (interference, scalability, QoS, security) in two sentences each.
6. For DTN: practice explaining the store-and-forward mechanism with a diagram showing two contact windows.
7. For VANET: list at least five ITS applications and the type of communication (V2V or V2I) each relies on.
8. Prepare one-page handwritten summaries of:
 - The full access network taxonomy with examples.
 - The infrastructure mode vs. ad hoc mode comparison.
 - Key properties of WSN, VANET, and DTN side-by-side.

4. Exercises

4.1. Access Networks and Wired Technologies

1. Draw a labeled diagram of a DSL access network showing the subscriber premises, telephone line, splitter, DSLAM, and ISP.
2. Explain how frequency division multiplexing is used in cable networks to separate video channels and data.
3. Compare DSL and cable networks in terms of technology, shared medium, and typical data rates.

4. What is the role of the edge router in an access network? Why is it called the “first router on the path”?

4.2. Wireless Network Components and Modes

5. Define the four components of a wireless network and give a real-world example for each.
6. Compare infrastructure mode and ad hoc mode. Give two examples of systems operating in each mode.
7. Fill in the four-quadrant classification table with at least two real system examples per cell.
8. A university deploys 802.11g access points throughout its campus. Which quadrant of the classification matrix does this represent? Justify your answer.

4.3. Single-Hop Infrastructure-Based Systems

9. Draw and label the 3G cellular data network architecture.
10. Explain the concept of cell reuse. Why does shrinking cell size increase capacity, and what are the trade-offs?
11. Tabulate the 802.11 WLAN standards (b, a, g, n, ac, ax, be) with their frequency bands and maximum data rates.
12. Explain how WiMAX serves as an alternative to DSL and cable for last-mile broadband access.
13. Distinguish between the roles of DSLAM in DSL networks and the base station controller in cellular networks.

4.4. Bluetooth and Single-Hop Ad Hoc Networks

14. Draw a Bluetooth piconet with one master and four slaves. Label two slaves as “Parked” and one as “Standby.”
15. Compare Bluetooth and 802.11 (WiFi) in terms of range, data rate, topology, and typical use cases.

4.5. Wireless Mesh Networks

16. Describe the roles of mesh routers and mesh clients in a WMN.
17. Explain the self-healing and self-configuration properties of WMNs with examples.
18. For each of the eight WMN challenges listed in the lecture, write one sentence explaining the problem and one sentence suggesting a mitigation strategy.
19. Describe two real-world applications of WMNs from the lecture (e.g., maritime, agricultural, smart city) and explain why a mesh topology suits each.

4.6. Ad-Hoc, Sensor, Vehicular, and Delay Tolerant Networks

20. Contrast a Wireless Ad-Hoc Network and a Wireless Mesh Network using at least five attributes.
21. List three characteristics shared by WSN, VANET, and DTN.
22. Explain why energy is the “primary constraint” in WSNs. How do sleep modes help, and what problems do they introduce?
23. Describe V2V and V2I communication with one practical example of each in an ITS application.
24. Explain Autonomous Intersection Management (AIM). What are its benefits over traditional traffic signals?
25. Explain the store-and-forward mechanism in DTNs.
26. Compare DTN with the traditional Internet in terms of delay, connectivity, and reliability mechanisms.

4.7. Comparison and Analysis

27. Construct a comparison table of Cellular Networks, WLANs, WiMAX, Bluetooth, and WMNs using columns: frequency, range, data rate, infrastructure, and primary use case.
28. Compare WSN, VANET, and DTN using the feature table from the lecture. Add a column for “primary application domain.”
29. List and briefly explain the four challenge categories of C2X communication (communication, networking, mobility, security).

5. Sample Questions

5.1. Short Questions

1. Define an access network. What is its role in the Internet architecture?
2. What is an edge router?
3. Name four types of access networks based on the technology used.
4. What is the difference between upstream and downstream transmission rates in DSL?
5. Define a base station and give two examples from the lecture.
6. What is a piconet in Bluetooth networking?
7. State two differences between infrastructure mode and ad hoc mode.
8. What is handoff in cellular networks?
9. Define co-channel interference in the context of cellular networks.
10. What does WiMAX stand for and what IEEE standard does it refer to?

11. What is the primary constraint in Wireless Sensor Network design?
12. Define “store-and-forward” in the context of DTNs.
13. What is V2V communication? Give one safety-related example.
14. List two differences between a mesh router and a mesh client in a WMN.
15. State two challenges of C2X communication.

5.2. Descriptive Questions

16. Describe the complete path a packet travels from a home computer connected via DSL to a web server on the Internet. Identify each network element and layer.
17. Explain the four-quadrant classification of wireless access networks (single-hop/multi-hop $\times$ infrastructure-based/infrastructure-less). Provide at least two examples in each quadrant.
18. Describe the 3G cellular data network architecture with a diagram. Explain the roles of the base station, radio access network, and MTSO.
19. Describe how a Wireless Mesh Network operates. Include the roles of mesh routers and mesh clients, multi-hop routing, self-healing, and self-configuration.
20. Explain the challenges of Wireless Mesh Networks. For any three challenges, suggest a potential solution or mitigation technique.
21. Compare Wireless Mesh Networks and Wireless Ad-Hoc Networks using at least six attributes. Justify which architecture is more suitable for a large-scale smart city deployment.
22. Describe the store-and-forward architecture of Delay Tolerant Networks. Why do conventional Internet protocols fail in DTN environments?
23. Explain Vehicular Ad-Hoc Networks (VANETs). Describe V2V and V2I communication and discuss at least three ITS applications enabled by VANETs.
24. Discuss the role of Wireless Sensor Networks in precision agriculture and military surveillance. How does the application domain affect the network’s structural organization (structured vs. unstructured)?

5.3. Analysis and Design Questions

25. A rural area with no existing cable or telephone infrastructure needs broadband connectivity. Analyze whether WiMAX or a Wireless Mesh Network is a more suitable solution. Justify using technical and practical criteria from the lecture.
26. A hospital wants to deploy a network to monitor patients’ vital signs in real time. Which type of wireless network (WSN, WMN, or WLAN) would you recommend? Justify your answer considering energy, latency, range, and reliability.
27. Consider a disaster-relief scenario in which communication infrastructure has been destroyed. Which type of wireless network should first responders deploy? Describe its operation and justify your choice.

28. Analyze the Autonomous Intersection Management (AIM) system: how does V2V/V2I communication eliminate the need for traffic signals?
29. Suppose you are designing a wildlife tracking system similar to ZebraNet. Identify the challenges specific to a DTN approach in this context and propose design choices to address each challenge.
30. Compare the suitability of Bluetooth, 802.11 (WiFi), and WiMAX for each of the following scenarios: (a) connecting peripherals to a laptop, (b) providing campus-wide Internet access, (c) providing broadband to a rural village. Justify each recommendation with reference to frequency, range, and data rate.
31. A city is deploying a smart transportation system with V2V-enabled truck platooning. Explain how V2V communication reduces brake lag, improves fuel efficiency, and enhances safety. Discuss the communication and security challenges that could undermine the system.
32. Evaluate the trade-offs between using a single-hop infrastructure-based wireless network versus a multi-hop infrastructure-less network for connecting 10,000 IoT sensors deployed across a 50 km² agricultural field.